

# Forside

## Eksamensinformation

AØKK08345E - Seminar: Anv. Corp Finance

## Besvarelsen afleveres af

Maria Louise Beathe Bardrum  
hjp130@alumni.ku.dk

## Eksamensadministratorer

### Bedømmere

Mikkel Godt Gregersen  
Eksaminator  
mikkel.godt.gregersen@econ.ku.dk

Jens Dick-Nielsen  
Censor  
jdn.fi@cbs.dk

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# Should Investors Consider CO<sub>2</sub> Efficient Investments?

Applied Corporate Finance

## An Empirical Analysis of the Performance and Relationship with CO<sub>2</sub> Efficient Portfolios

Supervisor Mikkel Godt Gregersen, December 2024

Maria Louise Beate Bardrum (hjp130)

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## Abstract

*This paper highlights the nuanced impact of sustainability metrics on portfolio performance, offering insights for climate-conscious investors.*

*This paper examines the relationship between CO<sub>2</sub> efficiency and portfolio performance in the European market from 2016 to 2024, utilizing the CAPM, Fama-French models, and Fixed Effects estimators. The analysis reveals a significant carbon risk premium, particularly for "brown" portfolios, low CO<sub>2</sub> efficiency, which achieve excess returns of 0.396 in CAPM and 0.380 in FF3 over the full period. Robustness tests of periods with low risk-free rates show that the greenest portfolios, high CO<sub>2</sub> efficiency, outperform, generating excess returns ranging from 0.0029 to 0.0035 in CAPM and 0.0034 to 0.0045 in FF3. Conversely, in more recent periods with higher risk-free rates, brown portfolios deliver the highest returns. These findings suggest that market dynamics and external factors, such as the Ukraine-Russia war, significantly influence investor preferences, favoring "safe" or high-emission investments. The hypothesis  $\alpha = 0$  can be rejected for all models at a minimum 10% significance level, indicating the presence of unexplained excess returns in the European market during the 2016 to 2024 period.*

*A deeper investigation into the relationship between returns and CO<sub>2</sub> efficiency was conducted using a fixed effects model. The analysis indicates a slight but insignificant negative relationship between CO<sub>2</sub> efficiency and returns, with a coefficient of  $-2.68 \times 10^{-9}$ . However, when high-emission sectors are excluded, a significant negative relationship emerges between Scope 1 emissions and annual returns, where a one-unit increase in emissions corresponds to a decrease of 0.001318% annual returns. This finding suggests that the carbon risk premium primarily arises in sectors where emissions are not essential to core operations, leading investors to demand higher returns as compensation for direct emission-related risks.*

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# 1 Introduction

The accelerating climate change and the implementation of stricter carbon regulations to limit emissions have increased uncertainty around carbon risk. This could mean that there is a potential risk for investors holding shares in companies with high emissions. Carbon risk may become a more significant factor for investors to consider, particularly if it evolves into a systematic risk factor. This would depend on the regulatory interventions, such as the implementation of a uniform carbon tax. As a result, the market might already be estimating the risks associated with holding these stocks different, which could lead to the existence of a carbon risk premium. According to modern portfolio theory, rational investors aim to achieve the highest possible return relative to the level of risk they are willing to accept. As sustainable investments are considered a limited subset of the overall investment opportunities. A restriction to sustainable investments could, in theory, negatively impact the risk-return balance.

Forward-looking investors may demand compensation for the risks associated with holding stocks in companies with high CO<sub>2</sub> emissions. Nonetheless, it is evident that many investors continue to allocate capital to companies with significant absolute emissions. This raises the question of how 'high CO<sub>2</sub> emitters' should be defined and investigated, as perspectives on what constitutes as harmful emissions can differ. For instance, should a bank with low carbon risk be considered a better investment compared to a large energy provider with substantial emissions? Or should the classification of high emitters be based on a company's efficiency and its efforts to minimize emissions relative to its revenue? To address these questions, I aim to explore whether the market rewards investors who prioritize emissions in proportion to revenue generated. Assessing if the most emission-efficient companies demonstrate evidence of generating abnormal returns and whether such a relationship exists.

**This paper seeks to examine if investors should consider CO<sub>2</sub> efficiency investments?**

The question addressed in this paper is investigated through a twofold approach. First, the Capital Asset Pricing Model (CAPM) and Fama French Models are used to determine if an abnormal return exists. Theoretically, no such premium should be present. *Hypothesis 1:  $\alpha = 0$* . Second, a relationship between carbon emission and stock return, does such relation even exists? A panel analysis can be conducted to examine this question. Here, the *Hypothesis 2: is that there is no significant relationship between CO<sub>2</sub> efficiency stock returns, meaning any coefficients would equal zero.*

The data used in this paper is derived from Refinitiv Eikon and the ECB Data Portal, covering the European market from October 2016 to September 2024. Key variables include CO<sub>2</sub> efficiency (calculated as CO<sub>2</sub> emissions relative to revenue), monthly returns, and portfolio performance relative to the STOXX Europe 600 index, and risk-free rates such as the Euribor 1-year rate.

**The results of the paper,** The findings of this paper, as highlighted in Sections 4, demonstrate that a carbon risk premium can be observed when portfolios are divided based on CO<sub>2</sub> efficiency. The time period under investigation plays a significant role in the results, as the analysis

of the entire period in Section 4 showed that the least efficient, or the brownest portfolio, generated an excess return of 0.396 in CAPM and 0.38 in FF3 model. And same results of the brownest portfolio in periods with a high risk-free rate, whereas during periods with a low risk-free rate, the most efficient portfolios outperformed, Q2, Q3, Q4 ranging from 0.0029-0.0045. A fixed effects estimator revealed an insignificant negative relationship between CO<sub>2</sub> efficiency and annual returns. However, when the analysis was extended by separating emissions into Scope 1 and Scope 2 categories and excluding all heavy-emission sectors, a significant negative relationship between direct emissions (Scope 1) and annual returns was identified.

## 1.1 Motivation

### *Scope Prioritizations*

Given the excessive focus on ESG reporting and its pillars, the motivation for this paper lies in examining the real impact and actual emissions, with CO<sub>2</sub> efficiency chosen as the primary variable of interest. Scope 1, direct emission and Scope 2 indirect emission are the impact indicators. Further definition of CO<sub>2</sub> efficiency will be provided in Section 3. Portfolios are constructed as equally weighted groups, divided into 25 pct. quartiles based on CO<sub>2</sub> efficiency. This paper focuses exclusively on Europe, as it represents the largest and a mature climate change oriented market (Management (2024)), utilizing data from Refinitiv Eikon. The analysis focus is to the Stoxx Europe 600 index, serving as a representative benchmark for the European market. Due to significant shifts in climate-related media focus in recent years and limitations in data quality as seen later in figure 1, historical data prior to 2015 is less relevant for this paper.

## 1.2 Literature Review

The concept of a carbon risk premium has been widely explored in the literature. As Friede et al. (2015) conducted a meta-study of over 2,000 published articles examining the relationship between corporate sustainability and financial performance. Their findings reveal that 90 pct. of the studies results vary depending on the methodology, geographic region, and asset classes. The following studies form the basis for addressing the central question of this paper.

*Some studies show that a green premium exists, but others studies show that investors maintain green portfolios despite earning lower returns.* Bernardini et al. (2021) identified a statistically significant Low-Carbon Premium, particularly pronounced since 2012. Where findings by Bolton and Kacperczyk (2021) that on an aggregate level Institutional investors tend to hold fewer shares in companies with high Scope 1 CO<sub>2</sub> intensity but find that that stocks of firms with higher total CO<sub>2</sub> emissions earn higher returns. Auer and Schuhmacher (2016) and Barber et al. (2021) finds, depending on the industry focus and the ESG criterion that is used, investors in Europe tend to pay a price for socially responsible investing. But Alessi et al. (2021) find that both firms' greenness and their level of environmental transparency are priced by the market. Their analysis reveals that greenhouse gas (GHG) emissions and the quality of environmental disclosures play a significant role in shaping investor decisions and show that investors are willing to accept lower returns to hold stocks that are greener and more transparent.

*Other studies indicate that no premium is observed.* Auer and Schuhmacher (2016) finds no ESG high rated stocks provide superior risk-adjusted performance in comparison to passive stock market investments. Görgen et al. (2017) examine the relationship between carbon risk and firm performance, identifying a significant ‘green premium’ for companies better prepared for the transition to a low-carbon economy, alongside a ‘carbon penalty’ for less prepared firms. Thus they observe two opposing effects: while brown firms tend to have higher average returns, decreases in the greenness of firms are linked to lower announcement returns. But they do not find evidence supporting the existence of a carbon risk premium.

*Premium depends on attention or legal framework,* Ramelli et al. (2021) finds the European stock price reactions to the global climate strike, that the event negatively affected the market valuation of carbon-intensive firms. And they found that the observed stock price penalty could be attributed to increased public attention to climate activism. Same conclusion is found by Monasterolo and De Angelis (2020) where following the Paris Agreement, the risk-return ratio for low-carbon companies decreased. Liang and Renneboog (2017) find that corporate climate-related considerations, are strongly correlated with the legal frameworks of the companies’ home countries.

As outlined above, the studies in this field reveal a landscape of conflicting conclusions.

## **2 Theoretical Foundation**

### **2.1 Portfolio theory**

To address the main question of this paper, modern portfolio theory (MPT) provides a foundational framework for understanding investment decisions. MPT posits that rational investors seek to maximize returns for a given level of risk Berk and DeMarzo (2017). While when grouping carbon effective investments in different portfolios, they are perceived as a subset of the total investment universe, potentially limiting diversification and impacting the risk-return trade-off.

Moreover, increased demand for corporate sustainable investments, could as suggested by Merton (1987) lead to sustainable portfolios underperformance compared to conventional investments due to higher pricing pressures. As an illustration, fewer investors may purchase brown stocks, due to carbon risk, causing their prices to drop despite offering higher returns, while the opposite may occur for green stocks, which become more expensive due to higher demand. So dividing portfolios into different CO<sub>2</sub> efficiency quantiles, thus different carbon risk could, in theory, negatively impact the risk-return balance. Within the CAPM framework, investors are compensated for systematic risks, as idiosyncratic risks are diversified away.

#### **2.1.1 CAPM and Jensen’s $\alpha$**

Capital Asset Pricing Model, (CAPM) will be used to investigate excess returns, the theory give insights on return and risk, thus estimating the expected return on an investment based on its risk. The model assumes investors are risk-averse hence when an investment has a higher

level of risk, investors demand a higher risk premium<sup>1</sup>. The CAPM is expressed by the following equation:

$$E[R_i] = r_f + \beta_i^{Mkt}(E[R_M] - r_f) + \varepsilon_{i,t} \quad (2.1)$$

where  $E[R_i]$  is the expected return of the investment  $i$ ,  $r_{f,t}$  is the risk-free rate and  $\beta_i^{Mkt}(E[R_M] - r_f)$  represents the sensitivity of the investment to market risk, where  $\beta = \frac{cov(r_i, r_m)}{var(r_m)}$ .  $\varepsilon_{i,t}$  is a measure of the investments volatility in relation to the overall market, thus in the theoretical world it is expected to be eliminated Berk and DeMarzo (2017).

Results of  $(\beta)$  values show a investments sensitivity to the market risk factor:  $\beta = 0$  indicates no sensitivity,  $\beta < 1$  low sensitivity,  $\beta = 1$  market-neutral,  $\beta > 1$  high sensitivity, and  $\beta < 0$  negative sensitivity.

A illustration of CAPM is the Capital Market line (CML) which focuses on the risk-return tradeoff for efficient portfolios, the Security Market Line (SML) complements this by depicting the relationship between an individual asset's systematic risk and its expected return, Figures can be seen in Appendix A.2. SML show the relationship between systematic risk in terms of  $\beta$  and the expected return. The SML is a straight line that illustrates the expected return for a given level of risk, thus if a stock or portfolio lies above the SML line, it indicates that it is undervalued or vica versa below/overvalued.

So if deviations occur from the SML, it indicates the presence of arbitrage opportunities, which contradict the assumptions of the CAPM. Under CAPM  $\alpha$  is expected to be zero, indicating no significant deviation from the predicted return. Jensen's alpha measures the difference between the actual return of a portfolio and the expected return calculated using the CAPM formula. By rearranging 2.1 and introducing Jensen's alpha as the intercept, the equation becomes:

$$E[R_i] - r_f = \alpha_i + \beta_i^{Mkt}(E[R_M] - r_f) \quad (2.2)$$

Where  $\alpha_i$  is the excess return which cannot be explained by systematic risks. If the alpha is positive, it indicates that the portfolio has generated an abnormal return and if negative indication the portfolio has underperformed.

The main idea of this paper is that a large part of the alpha is not explained by the factors included in models and the explanation for  $\alpha \neq 0$  should be attributed to carbon risk.

### 2.1.2 Fama-French 3-factor-model

In the scenario that stock returns may not solely depend on market risk, the CAPM model can be extended. Fama and French revealed that two additional factors, company size and value are key factors that significantly contribute to explaining variations in stock returns. Their models is as following:

$$R_{i,t} - r_{f,t} = \alpha_i + \beta_i^{Mkt}(R_{M,t} - r_{f,t}) + \beta_i^{SMB} SMB_t + \beta_i^{HML} HML_t + \varepsilon_{i,t} \quad (2.3)$$

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<sup>1</sup>Further assumptions under CAPM can be seen in A.1 and will not be discussed further in this paper, but is fundamental to CAPM model.

where *SMB* "The Size Sensitivity" represents the average return of the small-minus-big factor, calculated as the difference between the average returns of three small company portfolios and the average returns of three large company portfolios at time  $t$ . The size of the companies is determined based on their market capitalization<sup>2</sup>. Fama and French include this factor in their model due to the observed negative relationship between firm size and returns, indicating that smaller companies tend to outperform larger companies Fama and French (1993); French (2023).

*HML* "The Value Sensitivity" is the "high minus low" factor, representing the difference between the returns of portfolios consisting of companies with high and low book-to-market ratios, respectively. Fama and French include the *HML* factor in their model due to the observed positive relationship between a company's book-to-market ratio and its average returns. This suggests that stocks with low price-to-book ratios tend to outperform those with high price-to-book ratios French (2023).

## 2.2 Econometric Theory

### 2.2.1 OLS-estimation

The performance analysis of portfolios employs OLS estimation to determine the coefficients for each variable in the CAPM and FF3 models. Since both CAPM and FF3 are linear models, OLS provides consistent estimates for the variable of interest, Jensen's alpha. The assumptions under which OLS is unbiased are discussed in the corresponding section ??.

### 2.2.2 Fixed Effect Model

The objective is to delve deeper into the relationship between returns and carbon risk factors. This requires the use of a Fixed effects estimation, Fixed effects models are suited for panel data, where multiple observations exist for the same entities over time. By controlling for time-invariant characteristics unique to each entity, the model isolates the impact of variables that change over time. Panel data often includes unobservable firm-specific factors that may influence both the dependent variable, returns, and the explanatory variables. These firm-specific effects can be addressed effectively using the fixed effects estimator ?. This approach enables an investigation into the time-varying factors that could shape returns while accounting for entity-specific characteristics that remain constant over time.

The model can be illustrated as:

$$R_{i,t} = \alpha_i + \theta_i + \gamma_i CO2_{i,t} + \theta_i X'_{i,t} + u_{i,t}, \quad u_{i,t} \sim IID(0, \sigma_u^2) \quad (2.4)$$

where  $R_{i,t}$  represents the annual return for each company,  $CO2$  are the vector of the variables  $CO_2$  efficiency, Scope 1 and Scope 2. Thus making  $\gamma_i$  the main coefficient of interest. If  $\gamma_i$  for the emissions variables—Scope 1, Scope 2 or  $CO_2$  efficiency—is significantly negative, this indicates that an increase of 1 unit emissions would result in a decrease of annual returns by the coefficients pct.-point. And opposed for positive values.  $\theta_i$  represents time-specific fixed

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<sup>2</sup>Market capitalization is calculated as the product of outstanding shares and share price, obtained from Eikon Refinitiv.

effects.  $\alpha$  represents the fixed effects, where the model allow the fixed effects to be correlated with the explanatory variables. This is acceptable because  $\alpha_i$  is removed through a demeaning transformation, which accounts for time-invariant variation. This ensures that the estimators are not biased due to this correlation.  $X'_{i,t}$  is a vector of control variables that includes financial factors potentially impacting the company's risk premium. Similar  $\theta_i$  show the sensitivity of control variables.

### 2.2.3 Unbiased Estimators

Linear regressions are unbiased if the following assumptions are satisfied: 1) the explanatory variables are uncorrelated with the error term ( $E[u_i|X_i] = 0$ ), 2) the model is correctly specified as linear in parameters, 3) there is no perfect multicollinearity, and 4) the error term has constant variance ( $Var(u_i|X_i) = \sigma^2$ ) and no autocorrelation.

Additionally, Fixed Effects (FE) models are further unbiased when time-varying explanatory variables are uncorrelated with the idiosyncratic error term ( $E[u_{it}|X_{it}] = 0$ ) and there is sufficient variation within entities over time. FE controls for time-invariant unobserved heterogeneity by removing it through demeaning:

$$y_{it} - \bar{y}_i = (\beta_1(X_{it} - \bar{X}_i)) + (u_{it} - \bar{u}_i)$$

Both models rely on the key assumption of exogeneity to ensure unbiased and consistent estimates.

This paper employs t-tests to evaluate the significance of the alpha estimates, under the assumption that the residuals are normally distributed. Given the number of observations and the law of large numbers Verbeek (2017), I assume this condition is satisfied. However, if the residuals are not normally distributed, the results may falsely appear significant. To address this potential issue, I apply Newey-West standard errors, which correct for heteroskedasticity and serial correlation. This method is widely used for financial time series data, providing more reliable significance levels and reducing the risk of biased standard errors.

Additionally, for the Fixed Effects (FE) model, I account for both entity-specific and time-specific effects to control for unobserved heterogeneity across firms and over time.

## 3 Data and Portfolio Setup

### 3.1 Data Statistics

The dataset used in this paper is provided by data from Refinitiv Eikon and ECB Data Portal, covering from October 2016 to September 2024.

The variable of interest **CO<sub>2</sub> Efficiency** needs to be calculated to determine this papers main question and high CO<sub>2</sub> Efficiency serves as a proxy for how 'Green' companies are. I use the vari-

ables provided by Refinitiv Eikon<sup>3</sup>: *Scope 1 emissions*, *Scope 2 emissions* and *Total Revenue*<sup>4</sup>. Refinitiv Eikon collects fiscal annually Greenhouse Gas (GHG) Emission Data for companies<sup>5</sup>. Refinitiv Eikon follows the GHG Protocol World Resources Institute and World Business Council for Sustainable Development (2004) for all emission classifications by type. Scope 1 direct GHG emissions from the company, scope 2 which is indirect GHG emissions from the company and lastly Scope 3 which encompass emissions from the company’s entire value chain. Refinitiv Eikon and customary for other terminals such as Bloomberg, Morningstar, etc. the to exclude Scope 3. This is likely due to Scope 3 emissions being tied to the value chain, which is partially beyond the company’s control, and the data for Scope 3 being too uncertain, as suggested by Insights (2023). Consequently, Scope 3 emissions are excluded from analysis and will not be investigated in this paper.

Therefore, the company’s CO<sub>2</sub> Efficiency year-over-year is given by:

$$\text{CO}_2\text{Efficiency} = \frac{\text{CO}_2 \text{ Emissions Total}}{\text{Total Revenue (EUR)}} \quad (3.1)$$

Where CO<sub>2</sub> Emissions Total is calculated as the sum of CO<sub>2</sub> Emission Direct, Scope 1 and CO<sub>2</sub> Emission Direct, Scope 2). When CO<sub>2</sub> in billions of tons is divided by revenue in euros, you get a value that represents how much CO<sub>2</sub> is emitted for each euro generated. A value of e.g. 00.000000401 indicates that 4.01 kg of CO is emitted for each euro of revenue generated by each company in that year.

To measure the performance of the portfolios relative to **the market**, I used the Stoxx Europe 600 index, which comprises 600 leading publicly traded companies from 21 European countries<sup>6</sup>. This index provides a broad representation of the European market, which seen from table 7 as the biggest sector representing industrials and the smallest Energy weights respectively 21.37 pct. and 3.2 pct. of the index, and serves as a common benchmark for euro-denominated stocks. The index data was obtained from Refinitiv Eikon and includes only actively traded stocks, which can lead to survivorship bias.

The figure 1 show the aggregated CO<sub>2</sub> emissions for Scope 1 (direct emissions) and Scope 2 (indirect emissions) from 2015 to 2024. Scope 1 emissions, represented by the blue line, show a steady decline over the period, indicating a consistent reduction in direct emissions by companies. In contrast, Scope 2 emissions, represented by the green line, remain relatively stable with minimal variation, suggesting limited progress in reducing indirect emissions.

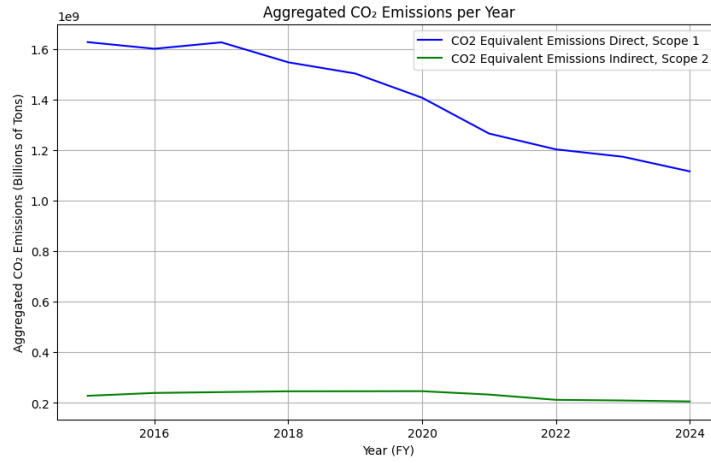
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<sup>3</sup>Eikon Refinitiv has its own Total CO2 equivalent variabel but it is given in USD and all my control variables and returns is in EUR, so I construct my own variable with the same method to avoid currency mix.

<sup>4</sup>which represents ‘*revenue from all of a company’s operating activities after deducting any sales adjustments and their equivalents*’ in Euros

<sup>5</sup>This variable represents the total emissions of carbon dioxide (CO<sub>2</sub>) and other greenhouse gases, measured in tonnes and the relevant gases include: carbon dioxide (CO<sub>2</sub>), methane (CH<sub>4</sub>), nitrous oxide (N<sub>2</sub>O), hydrofluorocarbons (HFCs), perfluorinated compounds (PFCs), sulfur hexafluoride (SF<sub>6</sub>), and nitrogen trifluoride (NF<sub>3</sub>).

<sup>6</sup>Austria, Belgium, Bermuda, Cyprus, Denmark, Faroe Islands, Finland, France, Germany, Ireland, Italy, Jersey, Luxembourg, Netherlands, Norway, Poland, Portugal, Spain, Sweden, Switzerland, United Kingdom

**Figure 1** Scope 1 and 2 Emissions Data for Stoxx Europa 600

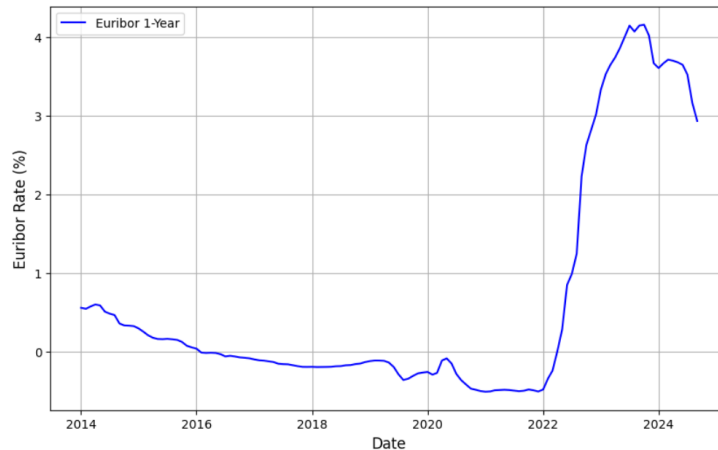
Source: from Refinitiv Eikon pr. Fiscal Year.

The table 1 shows the number of emission variables not reported for Stoxx Europa 600 index over time. The data indicates an increasing interest in reporting emissions, with a nearly 10 pct. decrease in missing values every two years, though the decrease slows from 2021 onwards. Therefore this paper drop missing observations where firms do not have data.

**Table 1** Missing CO<sub>2</sub> Emissions data in Stoxx Europa 600 index per year.

|                              | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 |
|------------------------------|------|------|------|------|------|------|------|------|------|------|
| NaN Count                    | 218  | 204  | 175  | 148  | 120  | 75   | 52   | 37   | 22   | 21   |
| Missing Value Percentage (%) | 37   | 34   | 30   | 25   | 20   | 13   | 9    | 7    | 4    | 4    |

To measure the performance of the portfolios, **the risk-free rate** is represented by the EURIBOR 1-year rate, obtained from the European Central Bank (ECB) Data Portal 2024. The recent evolution of the Euribor rate is particularly significant. Figure ?? shows the Euribor rate over the entire period. After a stable near-zero rate from 2016 to 2021, Euribor sharply increased in 2022 due to high inflation and tighter monetary policies, peaking above 4 pct. in mid-2023 before declining slightly (Lane (2024); Eurostat (2023)).

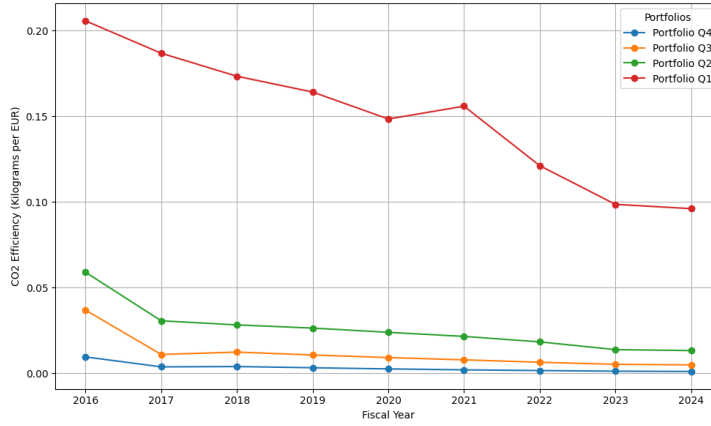
**Figure 2** Euribor 1-Year Rate (2014–2024)

### 3.2 Carbon-efficiency Equal Weighted Portfolios

To develop an passive investment strategy, I sort the data based on CO<sub>2</sub> emissions efficiency and dived into four 25 pct.-quantiles, to obtain a more nuanced view of the data. The first quantile, Q1, includes companies with the highest CO<sub>2</sub> efficiency absolute values, representing those with the lowest emissions efficiency per EUR revenue generated, thus a "Brown" portfolio. Conversely, the fourth quintile, Q4, contains companies with the lowest CO<sub>2</sub> efficiency values, indicating those that are most emissions-efficient companies, thus "Green" portfolio.

Figure 3 shows that the portfolios Q2, Q3 and Q4 are relative close to zero, where the most efficient portfolio Q4 in fiscal year 2024 with 1 gram<sup>7</sup> CO<sub>2</sub> emission per Euro spend. While portfolio Q1 demonstrates a decrease from 205 grams in fiscal year 2016 to 96 grams per EUR in 2024. This illustrates a significant difference in CO<sub>2</sub> efficiency between the least and most efficient companies.

**Figure 3** Average of CO<sub>2</sub> Efficiency per Fiscal Year for Each Portfolio



## 4 Analysis & Results of Carbon Efficiency

The analysis in this paper is structured into two sections to explore the impact of CO<sub>2</sub> efficiency. The first section 4.1 addresses *hypothesis 1*, examining whether excess returns are present in carbon-efficiency portfolios. The second section 4.2 investigates *hypothesis 2*, analyzing if a relationship between carbon emissions and stock returns exists.

### 4.1 CAPM and FF3

As described in 2, the return of a portfolio depends on the underlying risks of the assets it contains. All risk factors constructed and calculated from equation 2.1 and 2.3 and are following the methodology proposed by Fama and French<sup>8</sup> and reported estimates in table 2. To ensure the best unbiased estimator, I estimate the models using Newey-West standard errors to correct for heteroskedasticity and serial correlation in the residuals. I choose to use 4 lagged months,

<sup>7</sup>0.000990724413459122 kilograms per Euro

<sup>8</sup>Due to Fama and French only have the variables SMB and HML for SP500 I construct my own variables using Liabilities, Total Current Asset, Company market Capitalization obtained from Refinitiv Eikon .

representing one quarter, as these may influence the current residuals and help calculate robust standard errors.

**Table 2** CAPM and Fama-French 3-Factor Model, period 2016-2024

|                           | CAPM               |                     |                     |                     | FF3                |                      |                     |                       |
|---------------------------|--------------------|---------------------|---------------------|---------------------|--------------------|----------------------|---------------------|-----------------------|
|                           | Q1                 | Q2                  | Q3                  | Q4                  | Q1                 | Q2                   | Q3                  | Q4                    |
| $\alpha$                  | 0.396***<br>(3.37) | -0.011***<br>(-133) | -0.010***<br>(-121) | -0.010***<br>(-121) | 0.380***<br>(3.21) | -0.010***<br>(-123)  | -0.001***<br>(-111) | -0.011***<br>(-112)   |
| $\beta_i^{Mkt}$           | -0.030<br>(-0.04)  | 0.002***<br>(3.53)  | 0.002***<br>(3.33)  | 0.002***<br>(3.03)  | -0.073<br>(-0.10)  | 0.0058***<br>(12.0)  | 0.006***<br>(11.3)  | 0.0058***<br>(11.4)   |
| $\beta_i^{SMB}$           |                    |                     |                     |                     | 2.963<br>(0.69)    | -0.075***<br>(-26.2) | 2.1043<br>(-24.7)   | -0.0734***<br>(-26.1) |
| $\beta_i^{HML}$           |                    |                     |                     |                     | -2.865<br>(-0.70)  | 0.072***<br>(26.1)   | 0.072***<br>(24.7)  | 0.0704***<br>(25.9)   |
| <b>Adj. R<sup>2</sup></b> | 0,01               | 0,01                | 0,01                | 0,01                | 0,0                | 0,3                  | 0,3                 | 0,3                   |

Source: Eikon Refinitiv and own calculations. Note: The coefficients are estimations from equation 2.1 and 2.3 for the in CAPM and FF3 models.  $\alpha$  is in percentage per month. Levels of significance: \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ . t-test in parentes under the estimates, thus t-test helps reveal the impact of each coefficient, especially when standard errors are very small.

From table 2 it is shown that the excess return parameter,  $\alpha$  is significant for all the models. It is notable to see that only the  $\alpha$  is positive for Q1 portfolios, the brownest portfolio, in CAPM and FF3 models respectively on 0.396 and 0.380, similar positive excess returns, while the remaining portfolios are negative and around -0.011 to -0.001. This indicates that only Q1 delivers a potential risk-adjusted excess return beyond what the market factor can explain. This suggests that stocks of companies with high emissions but relatively lower revenue generation provide an excess return, given the associated market risk.

The relative similar estimations results observed in portfolios Q2, Q3, and Q4 could be attributed to their characteristics and portfolio composition, which appear to be closely aligned. As shown in Figure 3, the CO<sub>2</sub> efficiency of these portfolios is relatively similar, suggesting that their performance might be influenced by comparable underlying factors. This pattern is shown throughout the analysis.

#### *The sensitivity of the portfolios*

The market beta,  $\beta_i^{Mkt}$ , indicate relative minimal sensitivity to overall market movements under the CAPM model for Q2, Q3 and Q4, with coefficients near zero for all portfolios. Where for portfolio Q1 under both models Q1 retains a negative beta of -0.030 and -0.073 which means the portfolio moves in the opposite direction of the market but remains statistically insignificant in both models. For the portfolios of Q2, Q3 and Q4 in the Fama-French model, the inclusion of additional factors reveals significant positive market betas for Q2, Q3, and Q4 ranging from 0.0058 to 0.006, indicating moderate alignment with market dynamics.

The size factor,  $\beta_i^{SMB}$ , is significantly negative for Q2 and Q4, aligning with the expectation that the excess return is not generated by small companies, which may be true to the fact that the index contained for this paper include larger stocks from the European market. The Q1 and Q3 portfolios has a strong positive value but is insignificant. The value factor,  $\beta_i^{HML}$ , is

positive and significant for Q2, Q3 and Q4 portfolios under the Fama-French model, suggesting that value stocks contribute positively to the excess returns of these portfolios. Q1 portfolio is negative indicating that growth stocks contribute to excess returns but is insignificant.

It is notable that the explanatory power of the CAPM model is nearly 0 pct., indicating that the estimates may lack robustness with only a single market factor. Although I have accounted for heteroskedasticity and serial correlation in the residuals by estimating Newey-West standard errors. However, when additional explanatory factors are introduced in the 3-factor model, the explanatory power increases to 30pct. for portfolios Q2, Q3, and Q4. Interestingly, the explanatory power does not improve for the brownest portfolio, suggesting that the additional factors in the 3-factor model do not significantly capture the returns of this portfolio, potentially due to other unexplained characteristics specific to high-emission companies. Further robustness tests must be conducted.

#### 4.1.1 Do Time Periods Affect the Results?

As highlighted in 3 the risk-free rate has shown significant volatility in recent periods. To explore this, I estimated the CAPM and factor models across two distinct time intervals.

Table 3 presents the excess returns for the two divided periods. During the *period 2016–2022*, all portfolios exhibit positive and significant excess returns, which cannot be fully explained by CAPM and FF3. Portfolio Q4, the greenest portfolio, performs the best in both models, with coefficients of 0.0035 and 0.0045, respectively. In contrast portfolio Q1 show the smallest positive excess return in the same period. This could indicate that during times of low risk free rates in the period 2016–2022, investors may have increasingly favored greener portfolios.

In the *period 2022–2024*, a different pattern emerges. Significant and positive excess returns are observed only for portfolio Q1, the brownest portfolio, with coefficients of 0.8001 and 1.1458 (!). And Portfolios of Q2, Q3 and Q4 show negative relative close to zero ranging from -0.036 to -0.0268, indicating underperformance during this period. This suggests that investors may prioritize 'safe' investments during times of crisis, reflecting shifting preferences. This shift could also be attributed to the focus on the Ukraine-Russia war, where investors may have concentrated more on the energy market and heavy industries. The risk factor  $\beta$  for the two divided periods is shown in Appendix A.5.

**Table 3**  $\alpha$  in CAPM and Fama-French 3-Factor Model, divided periods

|                      | CAPM                |                     |                      |                      | FF3                 |                      |                     |                      |
|----------------------|---------------------|---------------------|----------------------|----------------------|---------------------|----------------------|---------------------|----------------------|
|                      | Q1                  | Q2                  | Q3                   | Q4                   | Q1                  | Q2                   | Q3                  | Q4                   |
| $\alpha^{2016-2022}$ | 0.0028***<br>(29.5) | 0.0029***<br>(33.2) | 0.0034***<br>(36.4)  | 0.0035***<br>(38.1)  | 0.0034***<br>(34.9) | 0.0034***<br>(39.0)  | 0.0040***<br>(41.7) | 0.0045***<br>(43.8)  |
| $\alpha^{2022-2024}$ | 0.8001***<br>(38.1) | -0.027***<br>(-245) | -0.0268***<br>(-236) | -0.0269***<br>(-230) | 1.1458***<br>(2.77) | -0.0365***<br>(-236) | -0.0364**<br>(-218) | -0.0364***<br>(-213) |

*Source: Eikon Refinitiv and own calculations.* Note: The coefficients are estimations from equation 2.1 and 2.3 for the in CAPM and FF3 models.  $\alpha$  is in percentage per month. Levels of significance: \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ . t-test in parentes under the estimates, thus t-test helps reveal the impact of each coefficient, especially when standard errors are very small.

The table in Appendix A.5 presents adjusted  $R^2$  values. For the 2016-2022 period, the FF3 model demonstrates higher explanatory power compared to the CAPM, with adjusted  $R^2$  values ranging from 0.08 to 0.09 across all portfolios. This indicates that, during this period, adding more explanatory variables was more relevant for explaining returns. In contrast, for the 2022–2024 period, the explanatory power of both models decreases. During this time, the CAPM occasionally outperforms the FF3 model, suggesting that the additional factors in the FF3 model (size and value) may lose relevance, potentially due to economic shocks such as the Ukraine war and high inflation. Notably, for portfolio of Q1 in the 2022-2024 period,  $\alpha$  reaches 0.80 in the CAPM and 1.45 in the FF3 model, indicating a strong excess return that cannot be explained by market, size, or value factors. However, the explanatory power ( $R^2$ ) for this portfolio is close to zero, which may explain the extreme  $\alpha$  value, as the models struggle to capture the variation in returns for Q1. This result may suggest a pattern rather than a consistent return. The explanatory power will be further discussed in Section 6.

Overall, the hypothesis  $\alpha = 0$  can be rejected at a minimum 10-pct. significance level for all models and periods analyzed in this paper. When this results show that  $\alpha \neq 0$ , it indicates that the portfolios exhibit an excess return that cannot be explained by the market, size or value factors are included in the models. Thus the results indicate a carbon risk premium in some periods as aligned with findings of Bernardini et al. (2021).  $\alpha$  is highest for the brown portfolios across the entire period. In the period with low risk-free rates, the models indicate that the greenest portfolio performs best, whereas in recent years, it is the brownest portfolio that outperforms in both CAPM and FF3 models.

## 4.2 Fixed Effect Estimator

The models above in section 4.1 aimed to assess whether an abnormal return in CO<sub>2</sub> efficiency divided portfolios, could be explained by market risk and other risk factors. Deeper investigation into the relationship between carbon risk and stock returns is required to examine how emissions may influence stock performance.

The Fixed Effect estimator from equation 2.4, is examined, and as previously mentioned,  $X'_{i,t}$  is a vector of control variables that includes financial factors potentially impacting the company's risk premium. Which financial factor impact the company's risk premium? is a difficult task to answer, so in order to find to most important factors to include I follow Bolton and Kacperczyk (2021), Flader (2022) and Nationalbank (2022). The importance of identifying the relationship between the variables of interest lies in holding the effects of control variables constant. By using the fixed effects estimator, time-invariant unobserved factors are controlled for, allowing for a more accurate estimation of the relationship between the variables of interest. The vector of controls includes a group of firm-specific variables know to predict returns such as market cap, revenue, beta, P/E, current assets, debt, equity, and liabilities, further arguments of the chosen variables can be seen in Appendix A.3. The control variables estimates are shown i Appendix 10.

To account for potential dependencies in the data, the estimates are calculated using clus-

tered standard errors, which allow for the error terms to account for serial correlation and heteroskedasticity across firms and time. Fixed effects are included to control for firm-specific differences between companies and to capture year-specific effects, addressing temporal variations such as the COVID-19 pandemic, the Ukraine war, and other significant economic or geopolitical events during the observed period. The results from the Fixed Effects estimator are presented in Table 4, displaying the estimated outcomes for total CO<sub>2</sub> efficiency in column (1). To delve deeper, Scope 1 and Scope 2 emissions are analyzed separately to determine whether either contributes significantly to returns. So Scope 1 and Scope 2 emissions individually in columns 2 and 3, respectively. Since emissions data are only available on a fiscal year basis, annual returns are used as the dependent variable to align with the explanatory variables and ensure model a more unbiased coefficient estimates.

Table 4 presents the regression results estimating the relationship between CO<sub>2</sub> efficiency, CO<sub>2</sub> emissions and annual returns. Column (1) shows the results for CO<sub>2</sub> efficiency, with a small negative coefficient of  $-2.68 \times 10^{-9}$ , indicating that higher total emissions are associated with slightly lower annual returns. Column (2) focuses on direct Scope 1 emissions, with the biggest coefficient of  $-9.62 \times 10^{-6}$ , suggesting that higher direct emissions have a marginally negative relationship with annual returns. Column (3) examines indirect Scope 2 emissions, with a coefficient of  $-7.37 \times 10^{-9}$ , also showing a negative relationship. However, the results are statistically insignificant, indicating no strong evidence of a relationship between CO<sub>2</sub> emissions—whether efficiency, Scope 1, or Scope 2—and annual returns.

**Table 4** Estimation of Regression Results

|                            | (1)                       | (2)                       | (3)                       |
|----------------------------|---------------------------|---------------------------|---------------------------|
| CO <sub>2</sub> Efficiency | $-2.68e^{-9}$<br>(-0.578) |                           |                           |
| Direct Scope 1             |                           | $-9.62e^{-6}$<br>(-1.117) |                           |
| Indirect, Scope 2          |                           |                           | $-7.37e^{-9}$<br>(-0.694) |
| Observations               | 3400                      | 2718                      | 3243                      |
| Companies                  | 491                       | 479                       | 489                       |
| Adj. $R^2$                 | 0.212                     | 0.201                     | 0.193                     |

Source: Eikon Refinitiv, CO<sub>2</sub> Equivalent Emissions

Note: \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ . I show the t-test in parentes under the estimates, thus t-test helps reveal the impact of each coefficient, especially when standard errors are very small. The t-statistic tests whether a coefficient is significantly different from zero. It is calculated as:  $t = \frac{\text{Coefficient}}{\text{Standard Error}}$ . A higher absolute t-value provides stronger evidence against the null hypothesis, indicating that the variable likely has a significant effect on the dependent variable. Drop in observations indicate missing values from Eikon Refinitiv.

This paper’s findings contribute to the findings with those of Bolton and Kacperczyk (2021), as they do not identify a significant relationship between CO<sub>2</sub> Efficiency and stock returns. even though given this paper and Bolton and Kacperczyk (2021) have differences in the time period analyzed and the frequency of data, as this study uses annual data while they employed monthly data.

To further explore the results from Table 4, additional analysis and robustness checks are conducted to ensure the validity of the findings.

#### 4.2.1 Sectors Specific Effects?

A robustness check for this paper's results involve excluding a CO<sub>2</sub>-intensive sector. This approach is based on the idea that the relationship between CO<sub>2</sub> emissions and returns may primarily stem from sectors that are not expected to have high emissions but exhibit poor CO<sub>2</sub> efficiency. Appendix A.4 provides a list of sub-industries within the Industrials, Materials, and Energy sectors, which together account for 32.9 pct. of the Stoxx Europe 600 index, as shown in Table 7. These sectors were selected for exclusion in the robustness check.

Table 5 presents the results after excluding the Industrials, Materials, and Energy sectors from the analysis. A statistically significant risk premium for Scope 1 emissions emerges at the 5 pct. significance level, indicating a negative relationship between Scope 1 emissions and annual returns for the remaining non-heavy-emission sectors in the Stoxx Europe 600 index. Specifically, an increase of one unit in Scope 1 emissions corresponds to a decrease in annual returns by -0.001318 pct. This finding suggests that the carbon risk premium primarily arise from sectors where emissions are not inherent to core operations, leading investors to demand higher returns as compensation for direct emission-related risks in these industries. For CO<sub>2</sub> efficiency and Scope 2 emissions, the relationships remain negative but insignificant, similar to the patterns observed in Table 4. This suggests that the coefficient is nearing zero for the remaining sectors, indicating that indirect emissions from industries like banking, retail, and similar sectors have minimal influence on annual returns. This could reflect that these sectors face lower risks associated with indirect emissions.

**Table 5** Robustcheck Estimation of Regression Results

|                            | (1)                       | (2)                         | (3)                        |
|----------------------------|---------------------------|-----------------------------|----------------------------|
| CO <sub>2</sub> Efficiency | $-2.76e^{-9}$<br>(-0.729) |                             |                            |
| Direct Scope 1             |                           | $-1.32e^{-5**}$<br>(-2.152) |                            |
| Indirect, Scope 2          |                           |                             | $-1.65e^{-10}$<br>(-0.010) |
| Observations               | 2051                      | 1624                        | 1968                       |
| Companies                  | 299                       | 290                         | 298                        |
| Adj. $R^2$                 | 0.174                     | 0.127                       | 0.143                      |

Source: Eikon Refinitiv, CO<sub>2</sub> Equivalent Emissions

Note: \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ . The  $t$ -test values are shown in parentheses under each estimate. The  $t$ -statistic tests whether a coefficient is significantly different from zero, calculated as  $t = \frac{\text{Coefficient}}{\text{Standard Error}}$ . A higher absolute  $t$ -value provides stronger evidence against the null hypothesis, indicating a significant effect of the variable on the dependent variable. Drop in observations indicate missing values from Eikon Refinitiv.

As shown in Table 5, the number of observations drops significantly from 2718 to 1624 in the robustness check, potentially leading to greater uncertainty in the estimated coefficients. Furthermore, the removed observations are not randomly distributed but systematically affect

specific groups. The explanatory power of both models is illustrated in Appendix A.6, where a steeper slope indicates a stronger correlation between predicted and actual values. While the models demonstrate explanatory power ranging from 12 % to 21%, it does not capture all variation in Yearly Return. The control variables are insufficiently informative for more complex patterns, and Fixed Effects remove much of the variation, leading to clustering around 0.1. There is potential to enhance the model by incorporating additional control variables. However, as mentioned at the beginning of this analysis, similar methodologies and findings are presented in Bolton and Kacperczyk (2021) and Nationalbank (2022), which supports the validity of the results in this paper.

Despite that the purpose of this robustness check was to explore whether a relationship could be identified in non-heavy-emission sectors, and the significant negative link between Scope 1 emissions and annual returns confirms this.

## 5 Discussion and Results Reflections

This section will first discuss the theoretical and data-related framework used for this paper. Next, the discussion will address how the empirical results relate to the existing literature and the relevance of the paper’s findings.

### 5.1 The Theoretical and Data-related Choices in the Paper

#### Method Considerations

This study employs the CAPM and Factor Models to estimate portfolio performance, incorporating additional factors such as size and value to add more explanatory variables to capture returns beyond market risk. These models rest on critical assumptions as stated in appendix A.1, including perfect markets, absence of transaction costs, taxes, and short-selling restrictions and rational, risk-averse investors with homogeneous expectations. While these assumptions simplify modeling, they may not fully reflect real-world complexities, such as diverse investor strategies or climate-related investor preferences. .

Alternatives to these models include extending Factor Models with additional variables like profitability, investment, and momentum, often referenced as RMW, CMA, MOM, or employing Arbitrage Pricing Theory (APT), which incorporates macroeconomic factors such as inflation and GDP growth. While these approaches enhance explanatory power and may give a more flexible realistic model, they risk of overfitting and require precise identification of risk factors. Moreover, incorporating too many factors increases the risk of overfitting, potentially reducing the model’s ability to accurately predict future outcomes based on historical data Berk and DeMarzo (2017).

As shown in Section 4, the CAPM and FF3 models demonstrate relative low explanatory power. However, as Israel et al. (2020), suggests that extending the time horizon to up to five years can significantly improve the signal-to-noise ratio, enhancing the explanatory power of the models. At the same time, longer periods reduce the number of available observations in my data,

increasing standard errors and prediction variance, thereby reducing the precision of the estimates. This trade-off could explain the lower explanatory power observed in the models used in this study. Future research could explore ways to improve explanatory power by extending the time horizon or incorporating additional factors, but as discussed this is not suitable for the purpose of this paper. The decision to use a simpler Fixed Effects and OLS estimation was made to ensure both consistency and practical applicability. Similarly, limiting the Fama-French model to only three explanatory factors is justified. Wooldridge (2018) supports this approach, noting that simpler models often provide the best fit for reliable estimations. Considering the limitation of data availability restricted to only fiscal year observations for emissions, the inclusion of additional factors could further affect the results and contribute to unnecessary model complexity.

### **Limitations of my Data**

This paper uses CO<sub>2</sub> efficiency as a key metric to evaluate corporate sustainability, focusing on emissions relative to revenue. While this approach provides insights into operational efficiency, it does not account for absolute emission reductions or annual changes in CO<sub>2</sub>, which may better reflect a company's commitment to sustainability. Relying on relative efficiency also risks misclassification, as high-revenue companies with significant absolute emissions might appear "greener". Additional broadening the analysis to include additional ESG dimensions, such as social or governance factors, could offer different perspectives. However, the choice of CO<sub>2</sub> efficiency remains relevant given investor behaviour. As highlighted by Hong and Kacperczyk (2009), institutional investors tend to divest significantly from firms with high Scope 1 emissions intensity but do not base decisions on total emissions or emission reduction. This investor behaviour is reflected in continued investments in large emitters like companies Volkswagen, E.on and Mærsk, suggesting that if investors care, investors would prioritize how efficiently companies use fossil fuels rather than focusing on absolute emission levels. This implies that investors are not necessarily concerned about reducing their overall exposure to carbon emissions risks, reinforcing the importance of CO<sub>2</sub> efficiency as a decision-making metric.

This paper account for look-ahead bias due to the fiscal data for CO<sub>2</sub> efficiency. The look-ahead bias could occur if stock returns are linked to emission data unavailable to investors at the time, potentially distorting the relationship between emissions and returns. So given the Fiscal part the emission data is typically released gradually during the first months of year  $t$ . To address this, this paper uses annual CO<sub>2</sub> data for year  $t - 1$  to analyze monthly returns for year  $t$ , ensuring emissions data is available before returns are evaluated. This may mean that some monthly data might not fully reflect the emissions data. Since the variable of interest is only reported annually, increasing the frequency of the data could allow the models to incorporate more information and better align with the law of large numbers. However, it should be noted that higher-frequency data might also be influenced by more significant and numerous external factors compared to lower-frequency data. Given the preference for ensuring a bigger sample of data includes emissions information, this approach is a reliable method. Furthermore Bolton and Kacperczyk (2021) include emissions data for year  $t$  in a robustness test, they find consistent results with this approach.

This paper addresses potential data selection bias, which could influence the analysis due to the increasing focus on climate change concerns and media attention in recent years. The dataset spans only 7 years, which might be too narrow to capture long-term patterns. However, as shown in Table 1, extending the dataset further back would result in poorer data quality. Additionally, older data may not accurately reflect current conditions, as highlighted by the argument that even with decades of data, estimates often exhibit large standard errors and fail to capture present market expectations Berk and DeMarzo (2017). The selected time period is therefore considered appropriate to balance data quality and relevance. A similar argument applies to the issue of survivorship bias, as the dataset includes only surviving entities, which could inflate performance. Nonetheless, given the relatively short observation period, this bias is unlikely to have a significant impact on the results.

## 5.2 Results Reflections

Reflecting on the results in comparison to the literature mentioned in section 1 highlights the varying nature of the findings and emphasizes the need for further investigation in this area to draw more definitive conclusions. This highlights the contribution of this paper, as it provides additional insights and further extends the topic. The findings in Sections 4 suggest that green portfolios generate excess returns during favorable economic periods, while direct emissions from sectors not inherently CO<sub>2</sub>-intensive in their core operations have a significant impact on returns across the entire time frame.

The results of this paper do not align with the findings of Auer and Schuhmacher (2016) and Görgen et al. (2017), which could be attributed to differences in methodology and factors considered. For instance, Auer and Schuhmacher (2016) used ESG ratings, whereas this study relies on CO<sub>2</sub> intensity, potentially leading to divergent outcomes. Similarly, Görgen et al. (2017) constructed a carbon risk factor-mimicking portfolio to analyze carbon risk. While their results demonstrate that carbon risk effectively explains systematic return variation, they do not find evidence of a carbon risk premium. In contrast, this paper employs a 25-quantile portfolio approach to identify a carbon risk premium, finding it in some periods but not in others.

This paper’s findings align with those of Bolton and Kacperczyk (2021), as both studies do not identify a significant relationship between CO<sub>2</sub> efficiency and stock returns. Despite differences in study design—including the time periods analyzed (2005–2017 vs. 2017–2024), the sources of carbon emissions data <sup>9</sup>, and the inclusion of additional explanatory factors in this study—these results contribute to the robustness and comparability of the findings. In contrast, Nationalbank (2022) find that reducing CO<sub>2</sub> emissions by 1% can increase a company’s share value by approximately 6%. The divergence in results may stem from methodological differences: Nationalbank (2022) estimate the fundamental value of stocks using a dividend discount model and applies a log-transformation of CO<sub>2</sub> emissions, whereas this study relies on the most recent observed monthly returns.

As mentioned in the theory section, Merton (1987) highlights that increased demand for green

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<sup>9</sup>Bolton and Kacperczyk (2021) used data from seven main providers, including CDP, Trucost, and MSCI, whereas this study utilizes data from Eikon Refinitiv

stocks can drive prices higher, potentially leading to overvaluation. Therefore, it may be surprising that in this paper, during the period from 2016 to 2022, green portfolios outperformed, rewarding investors. This could be attributed to an increasing number of investors committing to net-zero emissions targets. However, as discussed, the results in this paper suggest time-sensitive and may reflect short-term market- and legal framework-trends, raising the question of whether the market will continue to reward green investors in favorable economic conditions. According to finance theory, excess returns driven by mispricing are expected to diminish over time as arbitrage opportunities are exploited. On the other hand, if the excess returns are due to risks not yet captured by existing risk factors, these returns may persist in the long term. For instance, Brammer et al. (2006) finds that investing in companies with low social performance can yield higher returns, as portfolios of stocks from "socially least desirable companies" generate significant abnormal returns. Similarly, an newer study Sample and Coauthor (2024) finds that trading volume and momentum still have explanatory power for stock returns. Moreover, the globally renowned findings of Fama and French introduced influential factors affecting returns. These examples indicate that multiple factors can influence returns, underscoring the complexity of the market.

For future research, it would be valuable to explore higher-frequency CO<sub>2</sub> efficiency data. Additionally, incorporating rolling window regressions could provide deeper insights by dividing the data into overlapping periods and estimating the model over time. This is particularly relevant, as the findings in this paper demonstrate variations in results across different time periods.

***The purpose of this paper*** was to investigate whether incorporating CO<sub>2</sub> efficiency as a factor could influence investors' portfolio choices and returns. The findings suggest that green investors achieve excess returns in more positive market periods, while green portfolios tend to underperform during uncertain times e.g periods with high risk-free rates. This pattern could be explained by the notion that, in favourable economic conditions, investors may seek alternative investment opportunities and vica versa in adverse economic times. However, as the random walk theory emphasizes, historical returns cannot reliably predict future performance or patterns. These results highlight the complexity of integrating sustainability factors into investment strategies while considering market dynamics. This paper addresses a small aspect of a large and emerging field. The findings are preliminary and should be interpreted with caution.

## 6 Conclusion

The findings of this paper, as highlighted in Sections 4, demonstrate that an carbon risk premium can be observed when portfolios are divided based on CO<sub>2</sub> efficiency. The time period under investigation plays a significant role in the results, as the analysis of the entire period showed that the least efficient, or the brownest portfolio, generated an excess return of 0.396 in CAPM and 0.38 in FF3 model. And same results of the brownest portfolio in periods with a high risk-free rate, whereas during periods with a low risk-free rate, the most efficient portfolios outperformed, Q2, Q3, Q4 ranging from 0.0029-0.0045. A fixed effects estimator revealed an

insignificant negative relationship between CO<sub>2</sub> efficiency and annual returns. However, when the analysis was extended by separating emissions into Scope 1 and Scope 2 categories and excluding all heavy-emission sectors, a significant negative relationship of -0.001318 pct. between direct emissions (Scope 1) and annual returns was identified.

This paper concludes that a carbon risk premium does not exist in the overall period, but exists during periods of favourable economic conditions, but diminishes in more challenging market environments. Furthermore, the analysis reveals a significant negative relationship between direct emissions and returns within sectors not traditionally associated with high carbon intensity, suggesting that investors penalize firms for emissions outside their expected industry norms.

## 7 Literature

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## A Appendix

### A.1 CAPM Main Assumptions

**Investor Behavior:** All investors are rational, risk-averse, and seek to maximize the expected utility of their wealth. They evaluate potential investments based on expected returns and variances.

**Risk-Free Borrowing and Lending:** Investors can borrow or lend unlimited amounts at a risk-free interest rate, allowing them to adjust their portfolios without restrictions.

**Perfect Markets:** There are no transaction costs, taxes, or restrictions on short selling. All assets are infinitely divisible, meaning investors can buy fractional shares.

**Homogeneous Expectations:** All investors have access to the same information and therefore have identical expectations of an asset's returns, variances, and covariances.

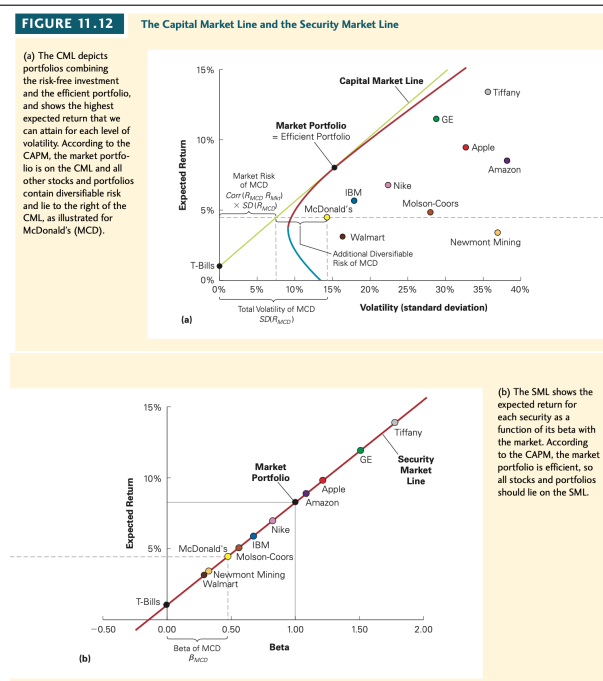
**Market Portfolio:** The market portfolio, which includes all risky assets, is assumed to be mean-variance efficient, meaning it provides the highest expected return for a given level of risk. Investors are assumed to always select the efficient portfolio, optimizing their risk-return balance.

**Investors are Price Takers:** Investors are small relative to the market, so individual buy or sell decisions do not affect asset prices.

**Single Period Investment Horizon:** Investors make decisions with a single-period horizon, meaning they are only concerned with the risk and return over a single time frame (e.g., one year).

### A.2 SML and CAPM

Figure 4 the Capital Market line and the Security Market line



Source: Berk and DeMarzo (2017) site 422 and 423

### A.3 Control Variables Arguments

**Table 6** Control Variables

| Control Variables                          | Arguments to include in the Fixed Effects model                                                                                                                                                                                                                                                                                                 |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Firm Size</i>                           | Measured as Company Market Capitalization. Larger companies may have more resources to invest in better CO <sub>2</sub> efficiency tools, which could influence stock returns.                                                                                                                                                                  |
| <i>Revenue</i>                             | Included in the model with a similar argumentation as Firm Size, reflecting resources and capacity that may influence CO <sub>2</sub> efficiency and stock returns.                                                                                                                                                                             |
| <i>Beta</i>                                | Captures a company's systematic risk, which determines its risk premium and stock returns. Controlling for beta ensures that market-driven risks do not confound the relationship between returns and carbon risk factors.                                                                                                                      |
| <i>P/E Ratio</i>                           | Captures time-varying firm-specific factors such as market perception and profitability that influence stock returns. By including it, the model isolates the effect of CO <sub>2</sub> -related variables while accounting for fluctuations in valuation over time, which can indicate the impact of CO <sub>2</sub> efficiency and emissions. |
| <i>Total Current Assets</i>                | Controls for the firm's liquidity and operational capacity, contributing to the understanding of financial stability in the model.                                                                                                                                                                                                              |
| <i>Total Debt, Equity, and Liabilities</i> | Ensures that the firm's financial health, stability, and structure are accounted for, controlling for their influence on CO <sub>2</sub> -related factors and stock returns.                                                                                                                                                                    |

Source: Refinitiv Eikon

### A.4 GICS Sector Name and belonging GICS Sub-Industry Name

Table 7 shows the breakdown of the STOXX Europe 600 index by GICS sectors. Industrials and Financials dominate with the highest number of companies, while sectors like Energy and Communication Services have smaller proportions.

**Table 7** Characteristics of STOXX Europe 600 per GICS Sector

| GICS Sector Name       | Number of Unique Companies | Proportion |
|------------------------|----------------------------|------------|
| Communication Services | 31                         | 5.2%       |
| Consumer Discretionary | 63                         | 10.5%      |
| Consumer Staples       | 46                         | 7.7%       |
| Energy                 | 19                         | 3.2%       |
| Financials             | 116                        | 19.4%      |
| Health Care            | 53                         | 8.9%       |
| Industrials            | 128                        | 21.4%      |
| Information Technology | 32                         | 5.3%       |
| Materials              | 50                         | 8.4%       |
| Real Estate            | 29                         | 4.8%       |
| Utilities              | 32                         | 5.3%       |

Table 8 highlights the sub-industries within the Industrials, Materials, and Energy sectors. It provides a detailed view of the industries, such as construction, chemicals, and oil gas, contributing to these broader sectors.

**Table 8** GICS Sub-Industries under Industrials, Materials, and Energy

| Industrials                                   | Materials                                      | Energy                             |
|-----------------------------------------------|------------------------------------------------|------------------------------------|
| Construction & Engineering                    | Steel                                          | Integrated Oil & Gas               |
| Marine Transportation                         | Specialty Chemicals                            | Oil & Gas Equipment & Services     |
| Electrical Components & Equipment             | Paper & Plastic Packaging Products & Materials | Oil & Gas Refining & Marketing     |
| Building Products                             | Diversified Chemicals                          | Oil & Gas Storage & Transportation |
| Air Freight & Logistics                       | Paper Products                                 | Oil & Gas Exploration & Production |
| Construction Machinery & Heavy Transportation | Industrial Gases                               |                                    |
| Heavy Electrical Equipment                    | Commodity Chemicals                            |                                    |
| Industrial Machinery & Supplies & Components  | Fertilizers & Agricultural Chemicals           |                                    |
| Aerospace & Defense                           | Construction Materials                         |                                    |
| Data Processing & Outsourced Services         | Aluminum                                       |                                    |
| Passenger Airlines                            | Copper                                         |                                    |
| Research & Consulting Services                | Metal, Glass & Plastic Containers              |                                    |
| Human Resource & Employment Services          | Forest Products                                |                                    |
| Industrial Conglomerates                      | Diversified Metals & Mining                    |                                    |
| Trading Companies & Distributors              |                                                |                                    |
| Security & Alarm Services                     |                                                |                                    |
| Airport Services                              |                                                |                                    |
| Environmental & Facilities Services           |                                                |                                    |
| Agricultural & Farm Machinery                 |                                                |                                    |
| Highways & Railtracks                         |                                                |                                    |
| Diversified Support Services                  |                                                |                                    |

## A.5 $\beta$ -values for the two periods

**Table 9**  $\beta$ -coefficients in CAPM and Fama-French 3-Factor Model, divided periods

|                     | CAPM                  |                        |                       |                       | FF3                 |                      |                       |                       |
|---------------------|-----------------------|------------------------|-----------------------|-----------------------|---------------------|----------------------|-----------------------|-----------------------|
|                     | Q1                    | Q2                     | Q3                    | Q4                    | Q1                  | Q2                   | Q3                    | Q4                    |
| <b>2014-2022</b>    |                       |                        |                       |                       |                     |                      |                       |                       |
| $\beta_i^{Mkt}$     | -0.0022***<br>(-4.29) | -0.0023***<br>(-4.97)  | -0.0023***<br>(-4.64) | -0.0026***<br>(-5.39) | 0.0019***<br>(3.66) | 0.0017***<br>(3.45)  | 0.0016***<br>(3.13)   | 0.0015***<br>(3.09)   |
| $\beta_i^{SMB}$     |                       |                        |                       |                       | 0.002<br>(0.05)     | 0.002<br>(0.79)      | 0.003<br>(1.02)       | -0.001<br>(-0.15)     |
| $\beta_i^{HML}$     |                       |                        |                       |                       | -0.007<br>(-0.29)   | -0.028<br>(-0.96)    | -0.004<br>(-1.18)     | -0.000<br>(-0.023)    |
| Adj. R <sup>2</sup> | 0,01                  | 0,01                   | 0,0                   | 0,0                   | 0,08                | 0,09                 | 0,08                  | 0,09                  |
| <b>2022-2024</b>    |                       |                        |                       |                       |                     |                      |                       |                       |
| $\beta_i^{Mkt}$     | 0.639<br>(0.45)       | -0.0231***<br>(-34.51) | -0.0234***<br>(-33.3) | -0.023***<br>(-32.4)  | -0.6345<br>(-0.18)  | 0.0119***<br>(9,25)  | 0.0122***<br>(8.75)   | 0.0123***<br>(8.65)   |
| $\beta_i^{SMB}$     |                       |                        |                       |                       | 8.507<br>(0.27)     | -0.091***<br>(-7.69) | -0.094***<br>(-7.38)  | -0.096<br>(-7.37)     |
| $\beta_i^{HML}$     |                       |                        |                       |                       | -0.95<br>(-0.036)   | -0.028***<br>(-2.96) | -0.0285***<br>(-2.71) | -0.0282***<br>(-2.62) |
| Adj. R <sup>2</sup> | 0,0                   | 0,07                   | 0,06                  | 0,06                  | 0,0                 | 0,1                  | 0,1                   | 0,1                   |

Source: *Eikon Refinitiv and own calculations*. Note: The coefficients are estimations from equation XXX in CAPM and FF3.  $\alpha$  is in percentage per month. Levels of significance: \*\*\* p<0.01, \*\* p<0.05, \* p<0.1. t-test in parentes under the estimates, thus t-test helps reveal the impact of each coefficient, especially when standard errors are very small.

## A.6 Scatter Plots of the Fixed Effects Estimator Results

Figure 5 Scatterplots of FE

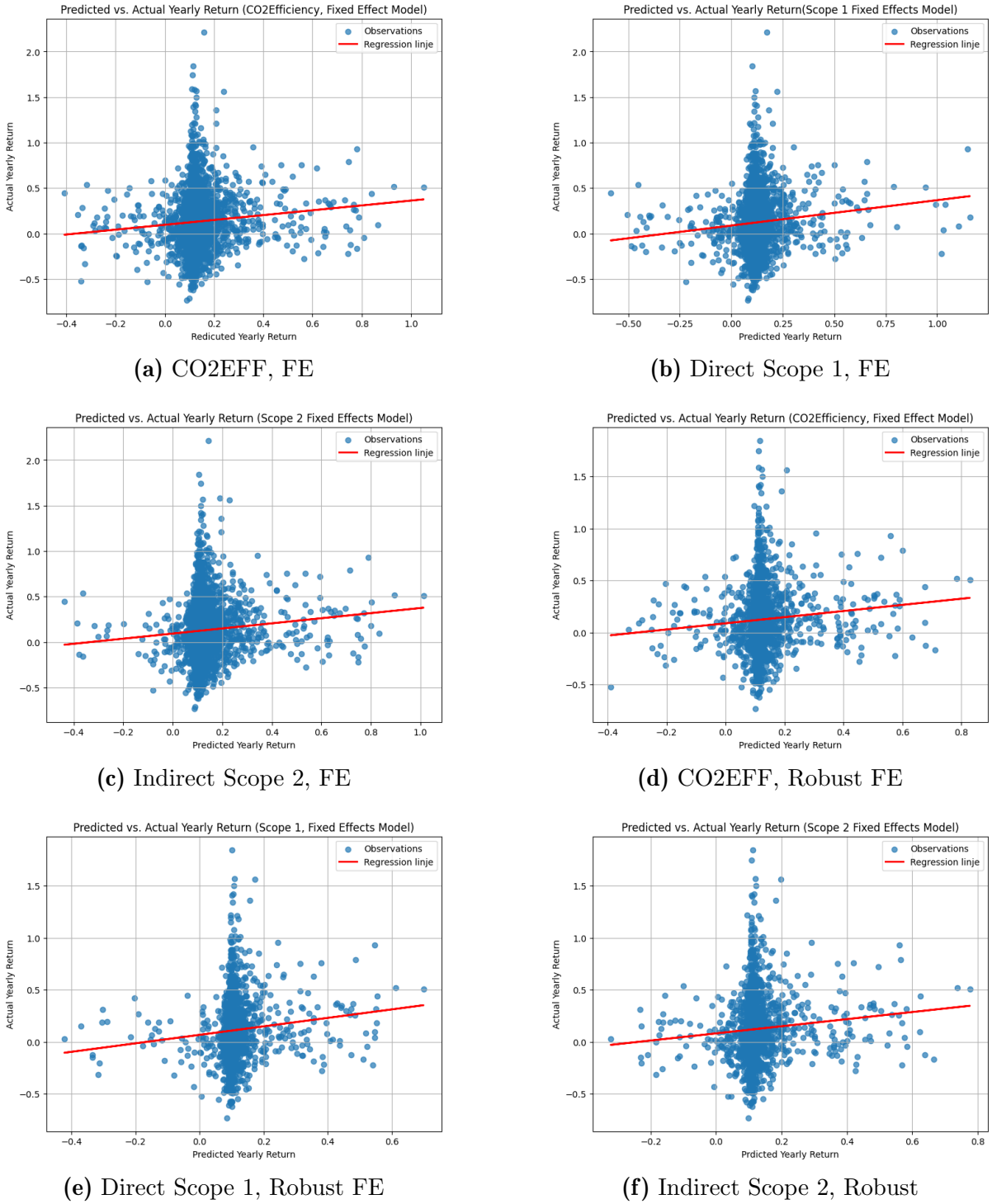


Figure 5 presents a series of scatter plots that depict the relationship between predicted and actual yearly returns under different specifications of the fixed effects model. Each panel corresponds to a different carbon metric—CO2 Efficiency, Direct Scope 1, and Indirect Scope 2—estimated using both standard and robust fixed effects methods. The scatter points represent observed data, while the red regression lines illustrate the average trend between predicted

and actual values. The explanatory power of both models is illustrated, where a steeper slope indicates a stronger correlation between predicted and actual values. While the models demonstrate explanatory power ranging from 12% to 21%, they do not capture all variation in yearly returns. The control variables are insufficiently informative for more complex patterns, and fixed effects remove much of the variation, leading to clustering around zero. There is potential to enhance the model by incorporating additional control variables.

## A.7 Control Variables from the Fixed Effect Estimations

**Table 10** Fixed Effect Estimator

|                               | (1)                           | (2)                           | (3)                           |
|-------------------------------|-------------------------------|-------------------------------|-------------------------------|
| CO <sub>2</sub> Efficiency    | $-2.689e^{-9}$<br>(-1.578)    |                               |                               |
| Direct Scope 1                |                               | $-9.62e^{-6}$<br>(-1.1166)    |                               |
| Indirect Scope 2              |                               |                               | $-7.373e^{-9}$<br>(-0.6941)   |
| Company Market Capitalization | $2.184e^{-12***}$<br>(4.1667) | $1.926e^{-12***}$<br>(3.3146) | $2.106e^{-12***}$<br>(3.8335) |
| Beta                          | $-0.0012$<br>(-0.0173)        | $-0.0110$<br>(-0.1357)        | $-0.0095$<br>(-0.1323)        |
| P/E (Daily Time Series Ratio) | $1.174e^{-5}$<br>(0.4997)     | $2.186e^{-6}$<br>(0.1424)     | $9.775e^{-6}$<br>(0.4458)     |
| Revenue                       | $1.284e^{-12}$<br>(0.9124)    | $2.546e^{-12}$<br>(1.9867)    | $1.339e^{-12}$<br>(0.9589)    |
| Total Current Assets          | $1.944e^{-12}$<br>(0.6675)    | $2.562e^{-12}$<br>(0.9125)    | $2.303e^{-12}$<br>(1.0023)    |
| Total Debt                    | $-2.788e^{-12}$<br>(-1.0962)  | $-4.014e^{-12}$<br>(-1.6664)  | $-2.261e^{-12}$<br>(-0.9178)  |
| Total Equity                  | $-3.604e^{-13}$<br>(-0.3172)  | $-3.077e^{-12}$<br>(-1.7984)  | $-5.186e^{-13}$<br>(-0.4078)  |
| Total Liabilities             | $-1.294e^{-12}$<br>(-0.5770)  | $-1.501e^{-12}$<br>(-0.6193)  | $-1.637e^{-12}$<br>(-0.7135)  |
| Observations                  | 3400                          | 2718                          | 3243                          |
| Companies                     | 491                           | 479                           | 489                           |
| Adj. $R^2$                    | 0.21                          | 0.20                          | 0.19                          |

Source: Eikon Refinitiv, CO<sub>2</sub> Equivalent Emissions

Note: \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ . The  $t$ -test values are shown in parentheses under each estimate. The  $t$ -statistic tests whether a coefficient is significantly different from zero, calculated as  $t = \frac{\text{Coefficient}}{\text{Standard Error}}$ . A higher absolute  $t$ -value provides stronger evidence against the null hypothesis, indicating a significant effect of the variable on the dependent variable.

**Table 11** Control Variables for the Robust check for Fixed Effects Estimator

|                               | (1)                         | (2)                           | (3)                          |
|-------------------------------|-----------------------------|-------------------------------|------------------------------|
| CO <sub>2</sub> Efficiency    | $-2.757e^{-9}$<br>(-0.729)  |                               |                              |
| Direct Scope 1                |                             | $-1.318e^{-5**}$<br>(-2.152)  |                              |
| Indirect Scope 2              |                             |                               | $-1.652e^{-10}$<br>(-0.010)  |
| Company Market Capitalization | $1.73e^{-12***}$<br>(4.331) | $1.373e^{-12***}$<br>(2.799)  | $1.626e^{-12***}$<br>(3.798) |
| Beta                          | $-0.0003$<br>(-0.005)       | $-0.0014$<br>(-0.018)         | $0.0003$<br>(0.005)          |
| P/E (Daily Time Series Ratio) | $7.47e^{-6}$<br>(0.315)     | $-4.269e^{-6}$<br>(-0.341)    | $5.3749e^{-6}$<br>(0.235)    |
| Revenue                       | $-8.73e^{-13}$<br>(-0.578)  | $3.91e^{-13}$<br>(0.258)      | $-8.886e^{-13}$<br>(-0.542)  |
| Total Current Assets          | $6.14e^{-12}$<br>(0.319)    | $-1.261e^{-12}$<br>(-0.484)   | $8.534e^{-13}$<br>(0.397)    |
| Total Debt                    | $-2.43e^{-12}$<br>(-1.098)  | $-5.437e^{-12**}$<br>(-2.099) | $-2.044e^{-12}$<br>(-0.816)  |
| Total Equity                  | $1.47e^{-12}$<br>(1.415)    | $6.279e^{-13}$<br>(0.414)     | $1.204e^{-12}$<br>(1.210)    |
| Total Liabilities             | $-1.88e^{-12}$<br>(-0.095)  | $1.701e^{-12}$<br>(0.651)     | $-4.197e^{-13}$<br>(-0.190)  |
| Constant                      | $0.1057^*$<br>(1.925)       | $0.0956$<br>(1.451)           | $0.1022^*$<br>(1.701)        |
| Observations                  | 2051                        | 1624                          | 1968                         |
| Companies                     | 299                         | 290                           | 298                          |
| Adj. $R^2$                    | 0.174                       | 0.127                         | 0.143                        |

Source: Eikon Refinitiv, CO<sub>2</sub> Equivalent Emissions

Note: \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ . The  $t$ -test values are shown in parentheses under each estimate. The  $t$ -statistic tests whether a coefficient is significantly different from zero, calculated as  $t = \frac{\text{Coefficient}}{\text{Standard Error}}$ . A higher absolute  $t$ -value provides stronger evidence against the null hypothesis, indicating a significant effect of the variable on the dependent variable.

Table 11 show Company Market Capitalization consistently demonstrates a positive effect at the 0.01 significance level across all three fixed-effect regressions. This suggests that as the market capitalization of a company increases, the annual returns of the company's stock are expected to increase, holding other factors constant. This may makes sense due to finance intuition.